

Assignment 4

CSE 350

$$(V - V_0) \left(\frac{C}{S} \right)_{\text{out}} = (V - V_0) \left(\frac{C}{S} \right)_{\text{in}}$$

Name: Rubyeat Wadud Galpe

$$(10.0(2.0 - 2.1) S)_{\text{out}} = (100.0(2.0 - 2.1) S)_{\text{in}}$$

ID: 24101376

$$20.2 \times \text{out} = \text{see 1 A}$$

$$20.2 \times \text{out} = 101 \times 15.1$$

$$101 \times 15.1 = \text{out}$$

$$1511.1 = \text{out}$$

CS

Ans to the ques no-1

$$A_{m1001.0} = \left[(100.0) - (100.0) \cdot 0.8 \right] 10.0 = 20.0$$

(a)

Case - I (Both inputs are High)

$$V_A = V_B = 5V$$

$M_1 = M_2 = \text{Triode}$

$$I_D = I_{DA} + I_{DB} \quad \text{--- (i)}$$

$$I_{DA} = 0.3 \times \left[2 \times (5 - 0.6) \cdot V_x - V_x^2 \right]$$

$$= 0.3 (8.8V_x - V_x^2) \quad \text{--- (ii)}$$

$$I_{DB} = 0.4 \left[2 \times (5 - 0.7) V_x - V_x^2 \right]$$

$$= 0.4 (8.6V_x - V_x^2) \quad \text{--- (iii)}$$

$$I_D = \frac{5 - V_x}{20} = 0.3 (8.8V_x - V_x^2) + 0.4 (8.6V_x - V_x^2)$$

$$\Rightarrow 5 - V_x = 6(8.8V_x - V_x^2) + 8(8.6V_x - V_x^2)$$

$$\Rightarrow 5 - V_x = 52.8V_x - 6V_x^2 + 68.8V_x - 8V_x^2$$

$$\Rightarrow 5 - V_x = 121.6V_x - 14V_x^2$$

$$\Rightarrow 14V_x^2 - 122.6V_x + 5 = 0$$

Solving this

$$V_x = 8.716V \quad \text{and} \quad V_x = 0.0409V$$

$$\therefore V_x = 0.041V$$

$$\therefore I_D = \frac{5 - 0.041}{20} = 0.248mA$$

$$\therefore I_{DA} = 0.3 [8.8(0.041) - (0.041)] = 0.1077 \text{ mA}$$

$$I_{DB} = 0.4 [8.6(0.041) - (0.041)] = 0.1404 \text{ mA} \quad (1)$$

here, (Both inputs are low) $V_x = 0.041 < 0.6 \text{ V}$.

$\therefore M_3 \rightarrow$ Cutoff mode

$$\therefore I_D = 0 \text{ mA}$$

$$\left[\frac{V_x - V_{th}}{R_{th}} - I_{G1} \right] \times 0.0 = A_0 I_D$$

$$\textcircled{II} \rightarrow \frac{5 - V_0}{20 \times 8.8} \times 0.0 = 0$$

$$\Rightarrow 5 - V_0 = 0$$

$$\textcircled{III} \rightarrow \frac{V_x - V_{th}}{R_{th}} \times 0.0 = A_0 I_D$$

Case - II

Both inputs are low

$$\left(\frac{V_x - V_{th}}{R_{th}} \right) \times 0.0 + \left(\frac{V_x - V_{th}}{R_{th}} \right) \times 0.0 = \frac{V_x - 2}{20} = I_D$$

$$V_A = V_B = 0 \text{ V}$$

$$\therefore M_1 = M_2 = \text{Cutoff}$$

$$\therefore I_D = I_{DA} = I_{DB} = 0 \text{ mA}$$

$$\therefore I_D = 0$$

$$\Rightarrow \frac{5 - V_x}{20} \times 0.0 = 0$$

$$\Rightarrow V_x = 5 \text{ V}$$

$$A_0 I_D = \frac{100.0 - 2}{20} = I_D$$

$\therefore M_3 \rightarrow$ triode.

$$I_o = 0.3 \{ 2(5-0.6)V_o - V_o^2 \}$$

$$\Rightarrow \frac{5-V_o}{20} = 0.3 \{ 8.8V_o - V_o^2 \}$$

$$\Rightarrow 5 - V_o = 6(8.8V_o - V_o^2)$$

$$\Rightarrow 6V_o^2 - 53.8V_o + 5 = 0$$

$$\therefore V_o = 8.87V \text{ and } V_o = 0.094V$$

$$\therefore V_o = 0.094V$$

$$\therefore I_o = 0.3 [8.8 \times 0.094 - (0.094)^2]$$

$$= 0.2453 \text{ mA}$$

(b)

Both inputs are high,

$$I_D = 0.248 \text{ mA}$$

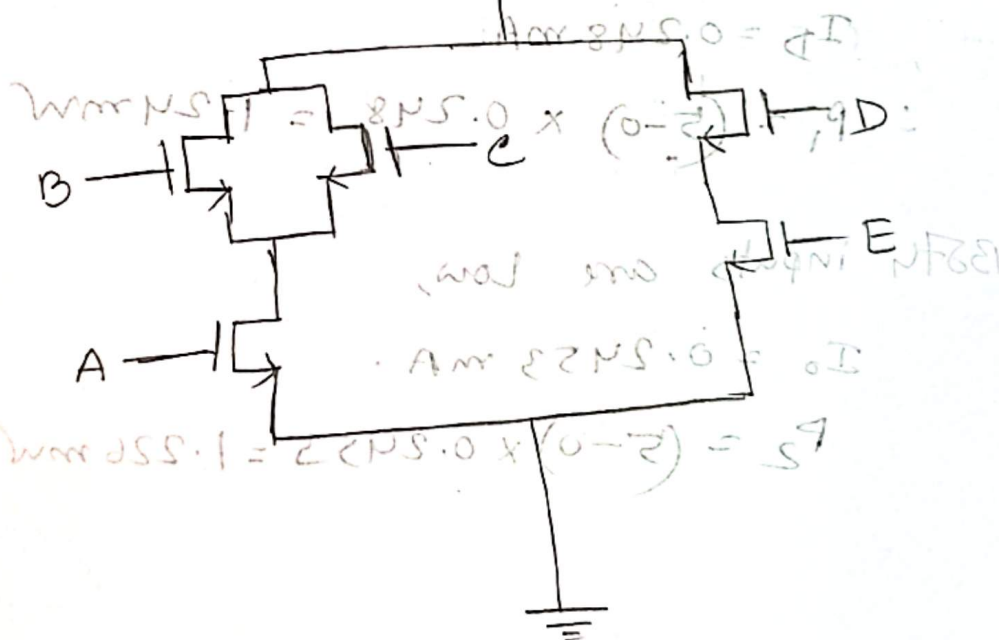
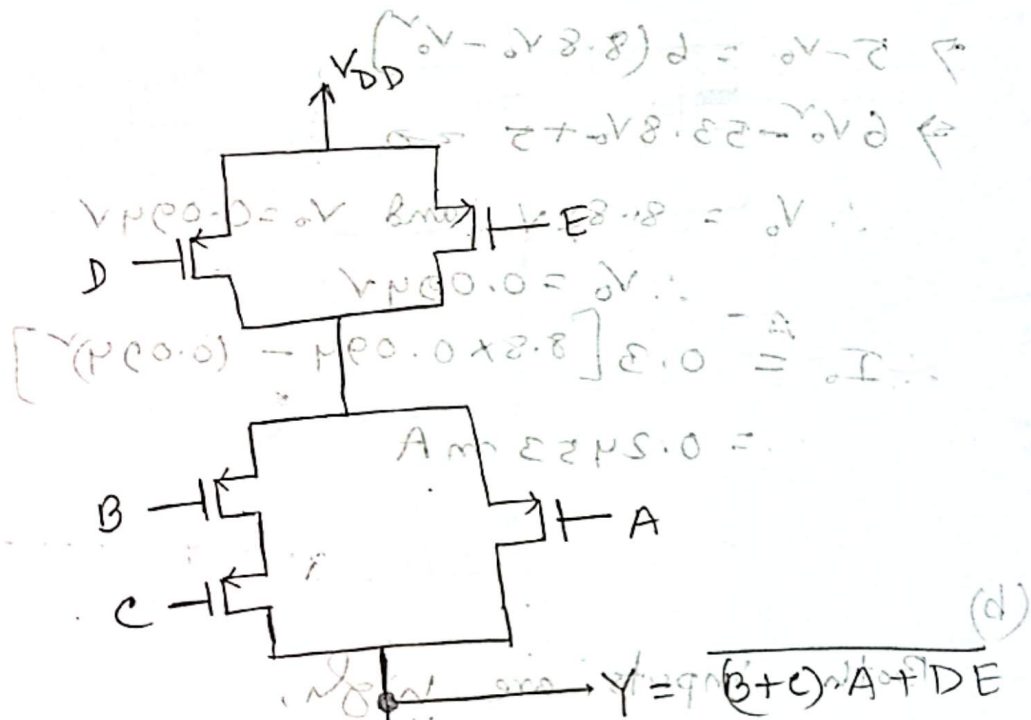
$$\therefore P_1 = (5-0) \times 0.248 = 1.24 \text{ mW}$$

Both inputs are low,

$$I_o = 0.2453 \text{ mA}$$

$$P_2 = (5-0) \times 0.2453 = 1.226 \text{ mW}$$

Boolean expression: $Y = \overline{(B+C) \cdot A + DE}$



Ans) - to the ques no-2

Wm sd. p.s.f = 150.2 mV x 2 = 1.9 (10000)

(a)

$$A = B = 5V$$

$T_1 \rightarrow$ reverse active

$$V_{R2} = 0V = 0V$$

$T_2 \& T_3 \rightarrow$ sat.

(11) - 5mV

$$\begin{aligned} \therefore V_{B1} &= V_{BE3} + V_{BE2} + V_{BC1} \\ &= 0.8 + 0.8 + 0.7 = 2.3V \end{aligned}$$

$$V_{20.0} = 1V$$

$$\therefore I_1 = \frac{6 - 2.3}{4.5} = 0.822 \text{ mA}$$

$$\beta_R = 0.2 \quad \text{Wm sd. p.s.f} = 10000$$

$$\begin{aligned} \therefore I_{B2} &= I_1 \times (1 + \beta_R) = 0.822 \times (1 + 0.2) \\ &= 0.9866 \text{ mA} \end{aligned}$$

Now,

$$V_{E2} = V_{B3} = V_{BE3} = 0.8V$$

$$I_q = \frac{0.8}{4} = 0.2 \text{ mA}$$

$$I_{E2} = I_q + I_{B3}$$

$$\Rightarrow I_{B3} = I_{E2} - 0.2$$

$$\Rightarrow I_{B3} = I_{B2} + I_2 - 0.2 \quad \left[I_{E2} = I_{B2} + I_2 \right]$$

$$\Rightarrow I_{B3} = 0.9866 + I_2 - 0.2$$

①

$$I_L = I_{B3} = 4.74 \text{ mA}$$

$$V_{C3} = V_{CE3} = 0.2 \text{ V}$$

$$\therefore I_3 = \frac{6 - V_{C3}}{4.5} = \frac{6 - 0.2}{4.5} = 1.2889 \text{ mA}$$

$$I_L = I_3 + I_{B3} = 4.74 - 1.2889 = 3.4511 \text{ mA}$$

$$V_{C2} = 5 - I_2 \times 2 = 5 - 2.6045 \times 2 = 2.6045 \text{ V}$$

$$V_{C2} = V_{CE2} + V_{CE3} = 0.8 + 0.2 = 1 \text{ V}$$

$$\therefore R = \frac{6 - V_{C2}}{I_2} = \frac{6 - 1}{2.6045} = 1.93 \text{ k}\Omega$$

$$R_{TH} = \frac{V_{C2}}{I_2} = \frac{1}{2.6045} = 19 \text{ }\Omega \approx 20 \text{ }\Omega$$

$$(b) \text{ P.P. } = \frac{1 - P}{5} = \frac{5 - 2}{5} = 0.6$$

Power

$$I_1 = \frac{5 - P}{5} = \frac{5 - 2}{5} = 0.6 \text{ A}$$

$$I_2 = 2.664 \text{ mA}$$

$$I_3 = 1.288 \text{ mA}$$

$$(0.6 \text{ A} + 2.664 \text{ mA} + 1.288 \text{ mA}) \times (0.2 \text{ V}) = 0.1288 \text{ W}$$

$$\therefore P = 6 \times (0.822 + 2.664 + 1.288) = 28.65 \text{ mW}$$

Answer to the ques no-3

(a) $V_s = 9V$

$A = B = \text{High}$. $\frac{9V - 2}{2.1} = 2I$

$T_1 \rightarrow$ Reverse active

$T_2 \rightarrow T_3 \rightarrow$ Saturation $I = 2I$

$V_{B1} = V_{BE3} + V_{BE2} + V_{BE1} = 0.8 + 0.8 + 0.7 = 2.3V$

$V_{C2} = V_{E2} + V_{CE2} = V_{BE2} + V_{CE2} = 0.8V + 0.2 = 1V$

$\therefore V_0 = V_{CE3} = 0.2V$

$I_{R1} = \frac{V_s - V_{B1}}{R_1} = \frac{9 - 2.3}{6.7} = 2mA$

$I_{R2} = \frac{V_s - V_{C2}}{R_2} = \frac{9 - 1}{2} = 4mA$

$I_{R3} = \frac{V_s - V_0}{R_3} = \frac{9 - 0.2}{4.4} = 2mA$

Power, $P = (V_s - 0) \times (I_{R1} + I_{R2} + I_{R3})$

$= 9 \times 7.4 = 66.6mW$

(b)

$$A = B = 0V$$

$T_1 \rightarrow$ forward bias | $T_2 = T_3 \rightarrow$ cutoff
 $\therefore I_{R_2} = I_{R_3} = 0mA$

$$V_{B1} = V_{IL} + V_{BE1} = 0 + 0.7 = 0.7V$$

$$I = \frac{V_S - V_{B1}}{R_1} = \frac{9 - 0.7}{R_1} = \frac{8.3}{R_1}$$

$$P = V_S \times I = 9 \times \frac{8.3}{R_1}$$

$$P < 1.23mW$$

$$\therefore \frac{74.7}{R_1} < 1.23$$

$$\Rightarrow R_1 > \frac{74.7}{1.23} = 60.73k\Omega$$

$$\therefore R_{min} = 60.73k\Omega$$

$$R_1 \in 60.73k, P = 1.23mW$$

The maximum value of $R_1 = \infty$ when $P = 0mW$

Ans to the ques no-4 (d)

$$\frac{0.064V}{0.001mA}$$

$$V_O = 0 = A$$

Q1

Lower threshold output when
 $V_A = V_B = 5V$

$$V_A = V_B \rightarrow \text{Triode } V_{DS} = V_{GS} = V_{DD} = 5V$$

$$\therefore I_D = I_{DA} + I_{DB}$$

$$\therefore \frac{5 - V_y}{20} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th})^2 \right] + \frac{1}{2} \mu_n C_{ox} \frac{W}{L} \left[(V_{GS} - V_{th})^2 \right]$$

$$\Rightarrow \frac{5 - V_y}{20 \times 10^{-3}} = 0.4 \left((5 - 0.25)V_y - V_y^2 \right) + 0.5 \times (5 - 0.3)V_y - V_y^2$$

$$\Rightarrow 5 - V_y = 800 (4.75 V_y - V_y^2) + 1000 (4.7 V_y - V_y^2)$$

$$\Rightarrow 1800 V_y^2 - 8501 + 5 = 0$$

Solving this $V_y = 0.064V$

$\therefore$ lower threshold output voltage = 0.064V

b

Case - I

$$V_A = V_B = 0V$$

$$I_R = 0mA$$

$$P_{1P} = 0mW$$

Case - II

$$V_A = V_B = 5V$$

$$V_y = 0.064V \text{ (from a)}$$

$$\therefore I = 0.2468mA$$

$$P_{2P} = 1.5 \times 0.2468 = 0.3702mW$$

Case - III

$$V_A = 0V \text{ \& } V_B = 5V$$

$M_n \rightarrow \text{cutoff}$ \& $M_p \rightarrow \text{triode}$

$$\frac{5 - V_y}{20 \times 10^3} = K_{n3} (2(V_{GS3} - V_{t3})V_{DS3} - V_{DS3}^2)$$

$$\Rightarrow 5 - V_y = 20 \times 10^3 \times 0.5 (2(4.7V - V_y) - V_y^2)$$

$$\Rightarrow 5 - V_y = 10000 (4.7V - V_y^2)$$

$$\Rightarrow \text{Solving this } \left. \begin{array}{l} V_y = 1.25mA \\ \therefore I = 0.1875mA \\ P = 5 \times I = 0.9375 \end{array} \right\}$$

Case - IV

$$V_A = 5V \quad V_B = 0V$$

MA + triode

$$V_{B1} = 0V = V_{B2}$$

$$A_{m0} = gI$$

$$\frac{5 - V_y}{20 \times 10^3} = 0.4 \times (5 - 0.25)V_y - V_y$$

$$\Rightarrow 5 - V_y = 800 (4.75V_y - V_y)$$

* Solving this

$$V_y = 1.24V$$

$$I = 0.188$$

$$P_{avg} = 5 \times 0.188 \times 1.24 = 0.94mW$$

$$\therefore P_{avg} = \frac{0 + 1.234 + 0.1375 + 0.94}{4}$$

$$V_2 = 0V \quad V_0 = AV$$

$$P_{avg} = 0.93mW$$

$$\frac{2 - V_y}{20 \times 10^3} = 0.4 \times (2 - 0.25)V_y - V_y$$

$$(2 - V_y) = 800 (1.75V_y - V_y)$$

$$(2 - V_y) = 1400 (0.75V_y - V_y)$$

$$A_{m0} = 0.188 = I$$

$$P_{avg} = I \times V = 0.188 \times 1.24$$

$$A_{m0} = 0.188 = I$$

(C)

$$V_A = V_B = 5V \quad | \quad M_A \rightarrow M_B \rightarrow \text{Triode}$$

$$I_{DA} = I_{De}$$

$$K_{NA}[(V_A - V_{TNA})V_Y - V_Y^2] = K_{NB}[2(V_B - V_{TNB})V_Y - V_Y^2]$$

$$\Rightarrow 0.5(2 \times 5 - 0.25) 0.64 \times 0.64 = K_{NC}(2(5 - 0.5)(0.64) - 0.64^2)$$

$$\Rightarrow 0.5(0.624 \times 4.096 \times 10^{-3}) = K_{NC} \times 5.35$$

$$\Rightarrow 1.27 \times 10^{-3} = K_{NC} \times 5.35$$

$$\Rightarrow K_{NC} = 2.37 \times 10^{-4}$$

$$= 0.237 \times 10^{-3} \text{ A/V}$$

$$\Rightarrow \text{scribbles}$$

Ans to the ques no-5

(a)

Case - I

$$(2.0 - V_A = V_B = 0V) \rightarrow \text{Cutoff}$$

$$(M_L \rightarrow \text{Linear}) \rightarrow \text{Linear}$$

$$I_D = 0$$

$$V_y = 5V$$

$$P_1 = (5V)(0A) = 0W$$

Case - II

$$V_A = 0V \text{ and } V_B = 5V$$

$$(M_A \rightarrow \text{Cutoff}, M_B \rightarrow \text{Linear}, M_L \rightarrow \text{saturation})$$

$$I_{DL} = I_{DB}$$

$$k_{nL}(V_{GS} - V_{TN})^2 = k_{nB}(2(V_{GS} - V_{TN})V_{DS} - V_{DS}^2)$$

$$\rightarrow 30(V_y - 0 + 0.6) = 10(2(5 - 0.5)V_y - V_y^2)$$

$$\rightarrow 30 \times 0.6 = 10(9V_y - V_y^2)$$

$$10V_y^2 - 90V_y + 10 = 0$$

$$V_y = 8.8V \text{ or } V_y = 0.12V$$

$$\therefore V_y = 0.12V \times 8.01 = 0.9612V$$

$$W_m: I_D = -30(0.12 - 0.12) + 0.6 = 10.8 \text{ mA}$$

$$P_2 = (5 - 0) \times 10.8 = 54 \text{ mW}$$

$$P_1 + P_2 + P_3 + P_4 = 9 + 27 + 54 + 9 = 109 \text{ mW}$$

Case - II

Same as Case - I

$$W_m: V_y = 0.12V \quad \left. \begin{array}{l} M_A \rightarrow \text{linear} \\ M_B \rightarrow \text{cutoff} \\ M_C \rightarrow \text{sat} \end{array} \right\}$$

$$P_3 = 54 \text{ mW}$$

Case - A

$$V_A = V_B = 5V$$

$$M_A = M_B = \text{triode}$$

$$M_C \rightarrow \text{sat}$$

$$I_{DL} = I_{DA} + I_{DB}$$

$$\rightarrow I_{DL} = 2 \cdot I_{DA}$$

$$\rightarrow 30(0.6) = 2 \times \left[10 \left\{ 2(5 - 0.5)V_y - V_y^2 \right\} \right]$$

$$\rightarrow 10.8 = 20 \times (9V_y - V_y^2)$$

$$\rightarrow 20V_y - 18V_y^2 + 10.8 = 0$$

$$\therefore V_y = 8.94V \text{ and } V_y = 0.0604V$$

$$\therefore V_y = 0.0604V$$

$$\therefore I_D = 10.8 \text{ mA}$$

$$P_4 = (5-0) \times 10.8 \text{ mA} = 54 \text{ mW}$$

$$P_{Diss} = 8.01 \times (5-0) = 40.05 \text{ mW}$$

$$P_{Total} = P_1 + P_2 + P_3 + P_4 = 0 + 54 + 54 + 54 = 162 \text{ mW}$$

$$P_{Avg} = \frac{162}{4} = 40.5 \text{ mW}$$

(b) Case-1
from (a)

$$\text{When } V_A = V_B = 0$$

$$V_Y = 5V$$

$$M_D \rightarrow \text{Linear}, M_{DL} \rightarrow \text{sat}$$

$$I_{DD} = I_{DL}$$

$$\rightarrow K_n (2(V_{GS} - V_t)V_{DS} - V_{DS}^2) = K_n (V_{GS} - V_t)^2$$

$$\rightarrow 50 (2(5-1)V_0 - V_0^2) = 10 (5-1-0.8)^2$$

$$\rightarrow 40V_0 - 50V_0^2 = 17.6 - 8.4V_0 + V_0^2$$

$$\rightarrow 6V_0^2 - 48.4V_0 + 17.6 = 0$$

$$\therefore V_0 = 7.68V \text{ and } V_0 = 0.38V$$

$$\therefore V_0 = 0.38V$$

$$\therefore I_{DL} = 10(4.2 - 0.38) = 145.924 \text{ mA}$$

$$\therefore \text{Power, } P_1 = 5 \times 145.924 = 729.62 \text{ mW}$$

Case - (ii)

$$V_A = V_B = 5V$$

from (a)

$$V_y = 0.66 V$$

$M_D \rightarrow \text{Cutoff}$ & $M_{L2} \rightarrow \text{sat}$

$$V_{E5} = 5.0 + 8.0 + 8.0 = 21.0$$

$$I_D = 0 \text{ mA}$$

$$P_2 = 0 \text{ mW}$$

$$P_{\text{Total}} = 729.62 \text{ mW}$$

$$(5.0 + 1) \times 558.0 = (5.0 + 1) \times 558.0 = 3546.0$$

$$A_{M2} = 0.8289.0 = 8289.0$$

$$V_{E5} = 5.0 + 8.0 + 8.0 = 21.0$$

$$A_{M5} = 0.8 \times 0.5 = 0.4$$

$$I_{E5} = I_D + I_{B3} = 0.5$$

$$I_{B3} = I_{E5} - 0.5$$

$$I_{B3} = I_{E5} - 0.5 = 0.5 - 0.5 = 0$$

$$I_{E3} = 0.5 + 0.5 = 1.0$$

①

Ans to the ques no-6

(Q)

V_{IT} : both M_n and M_p are in transition point.

for $M_p \rightarrow I_{DP}(sat) = I_{DP}(Triode)$

for $M_n \rightarrow I_{DN}(sat) = I_{DN}(Triode)$

$$\therefore I_{DP} = I_{DN}$$

$$\Rightarrow I_{DP}(sat) = I_{DN}(sat)$$

$$\Rightarrow K_p (V_{SGP} + V_{tp})^2 = K_n (V_{GSN} - V_{tn})^2$$

$$\Rightarrow K_p (V_{DD} - V_I + V_{tp})^2 = K_n (V_I - V_{tn})^2$$

$$\Rightarrow \sqrt{K_p} (V_{DD} - V_{It} + V_{tp}) = \sqrt{K_n} (V_{It} - V_{tn}) \quad [V_I = V_{It}]$$

$$\Rightarrow \sqrt{K_p} (V_{DD} + V_{tp}) - \sqrt{K_p} V_{It} = \sqrt{K_n} V_{It} - \sqrt{K_n} V_{tn}$$

$$\Rightarrow V_{It} (\sqrt{K_p} + \sqrt{K_n}) = \sqrt{K_p} (V_{DD} + V_{tp}) + \sqrt{K_n} V_{tn}$$

$$\Rightarrow V_{It} = \frac{\sqrt{K_p} (V_{DD} + V_{tp}) + \sqrt{K_n} V_{tn}}{\sqrt{K_p} + \sqrt{K_n}}$$

V_{ONT} : M_n is in transition point.

$$V_{DSN} = V_{GSN} - V_{tn}$$

$$\Rightarrow V_O = V_I - V_{tn}$$

at this point $V_O = V_{ONT}$

$$\therefore V_I = V_{IT}$$

(b)

$$V_{ONT} = V_{IT} - V_{tn}$$

V_{opt} : M_P is in transition point.

$$V_{DD} - V_0 = V_{DD} - V_I - |V_{tp}|$$

$$V_{DD} - V_0 = V_{DD} - V_I - |V_{tp}|$$

at this point $\rightarrow V_0 = V_{opt}$

$$V_{DD} - V_0 = V_{DD} - V_I - |V_{tp}|$$

$$V_{opt} = V_{IT} + |V_{tp}|$$

$$V_{IT} = \frac{\sqrt{10}(5-0.5) + \sqrt{10} \cdot 0.5}{\sqrt{10} + \sqrt{10}} = 2.5 \text{ V}$$

$$V_{IT} = 2.5 \text{ V}$$

$$V_{ONT} = 2.5 - 0.5 = 2 \text{ V}$$

$$V_{ONT} = 2 \text{ V}$$

$$2.5 - 0.5 = 2 \text{ V}$$

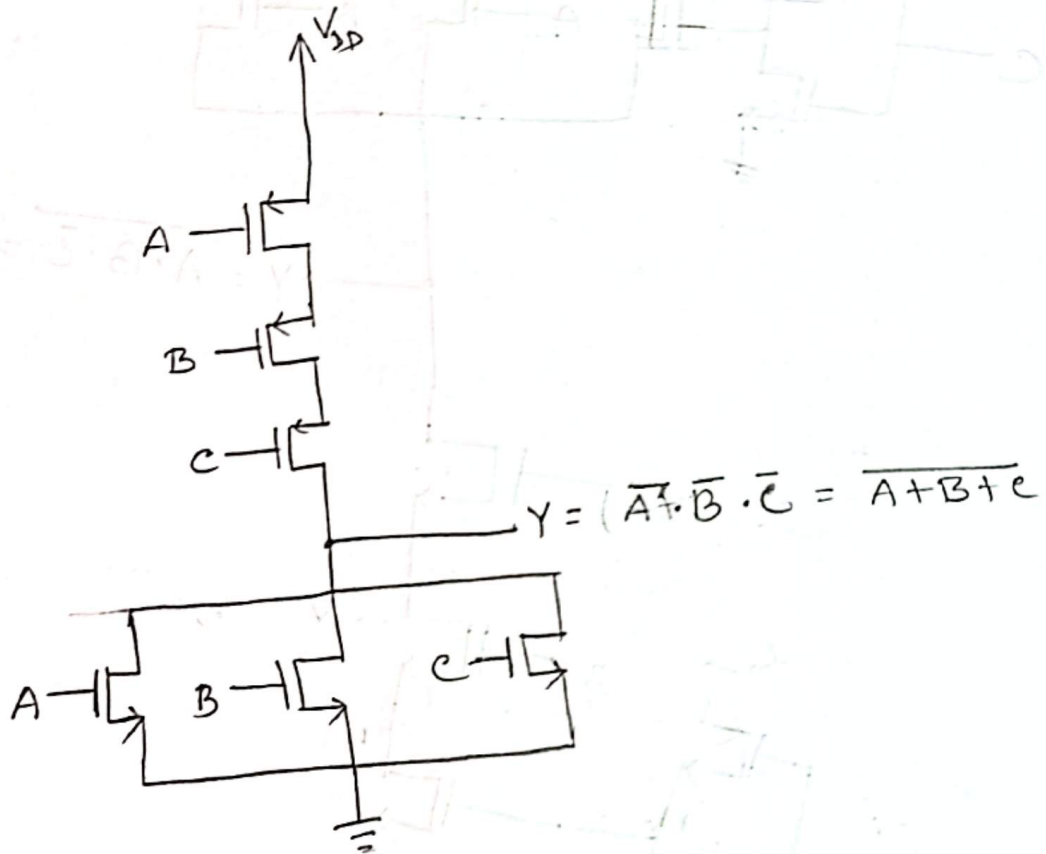
$$V_{opt} = 2.5 + 0.5 = 3 \text{ V}$$

$$V_{opt} = 3 \text{ V}$$

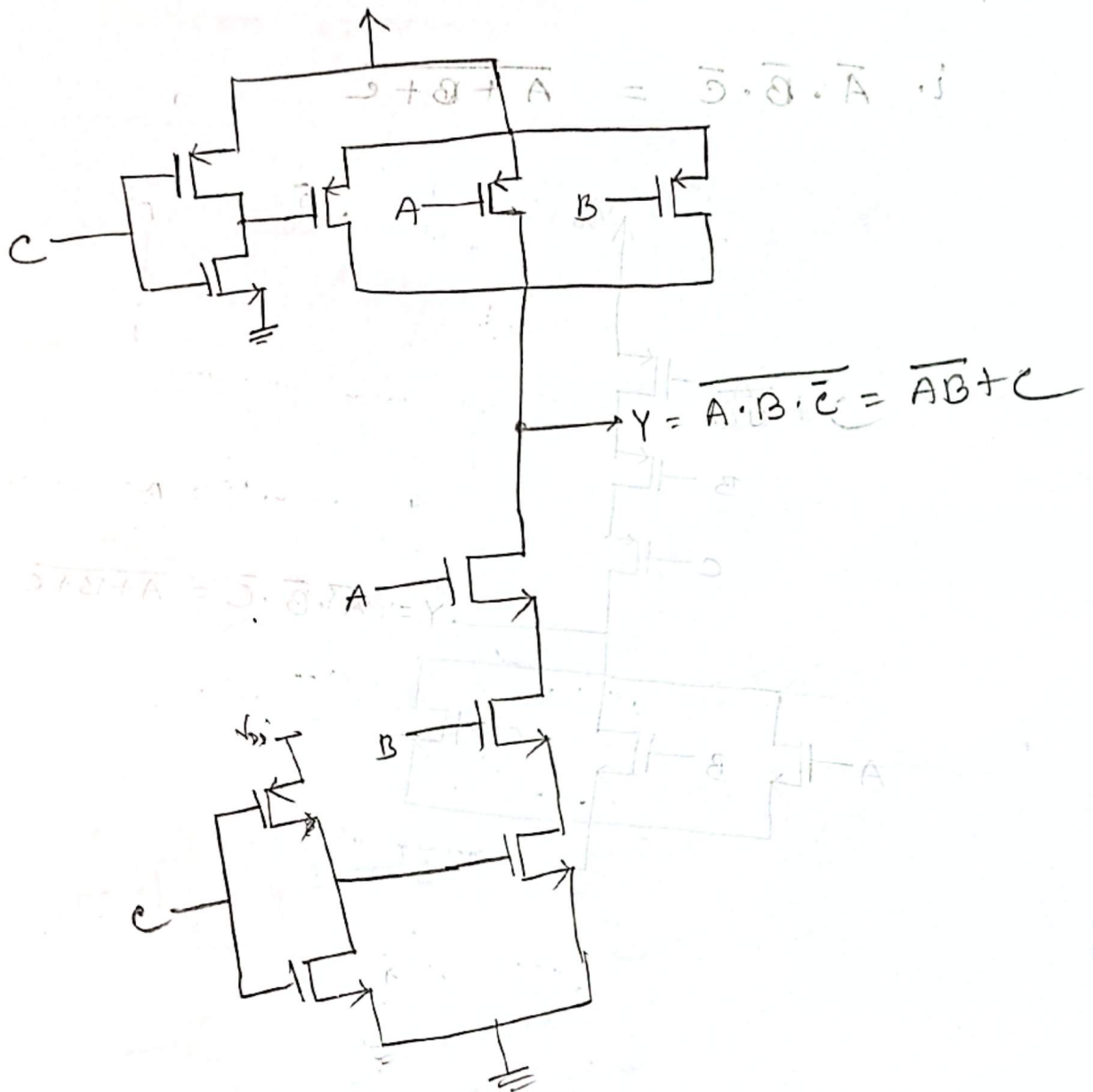
$$V_{opt} = 3 \text{ V}$$

5.8. Ans to the ques no. 7 + $\overline{BA}$ (ii)

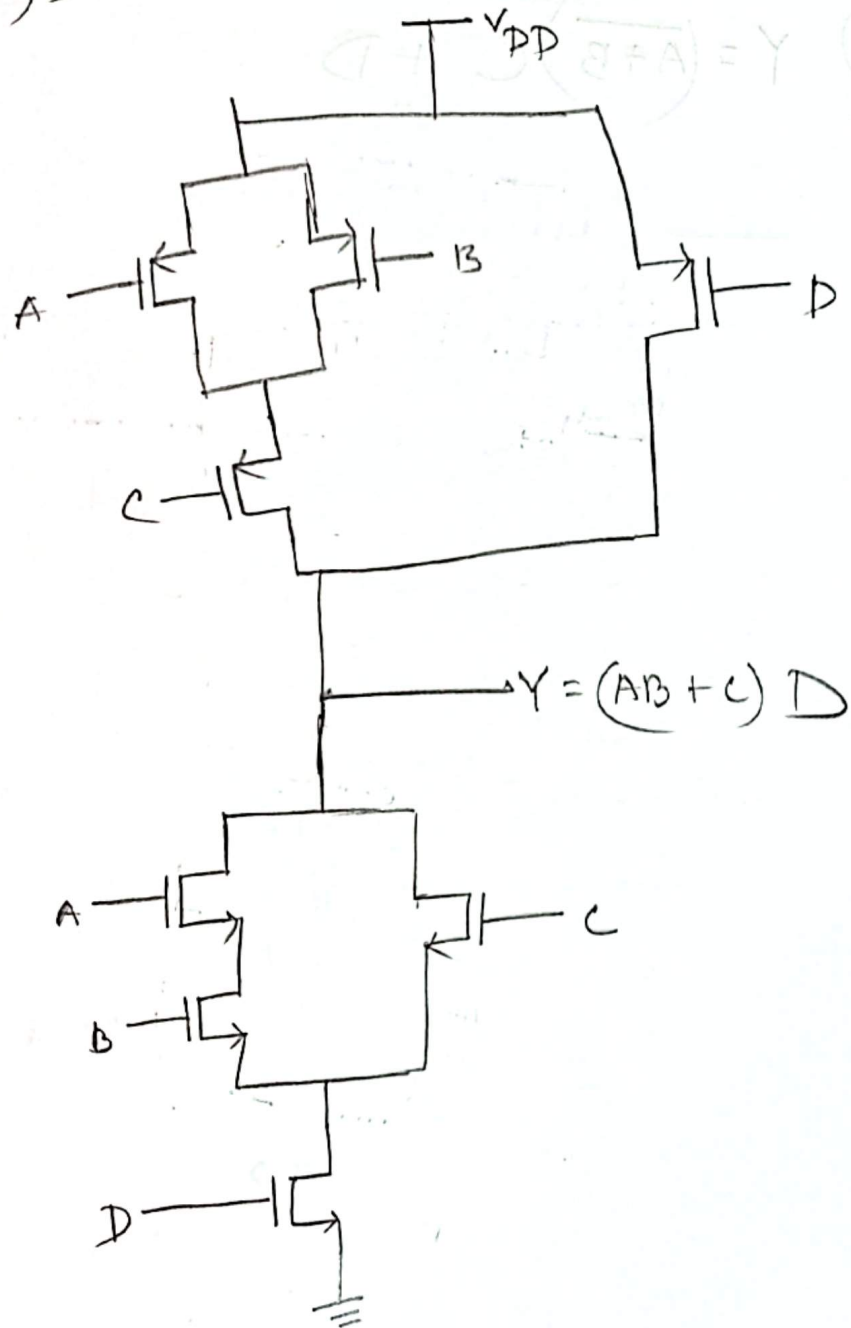
i. $\overline{A} \cdot \overline{B} \cdot \overline{C} = \overline{A+B+C}$



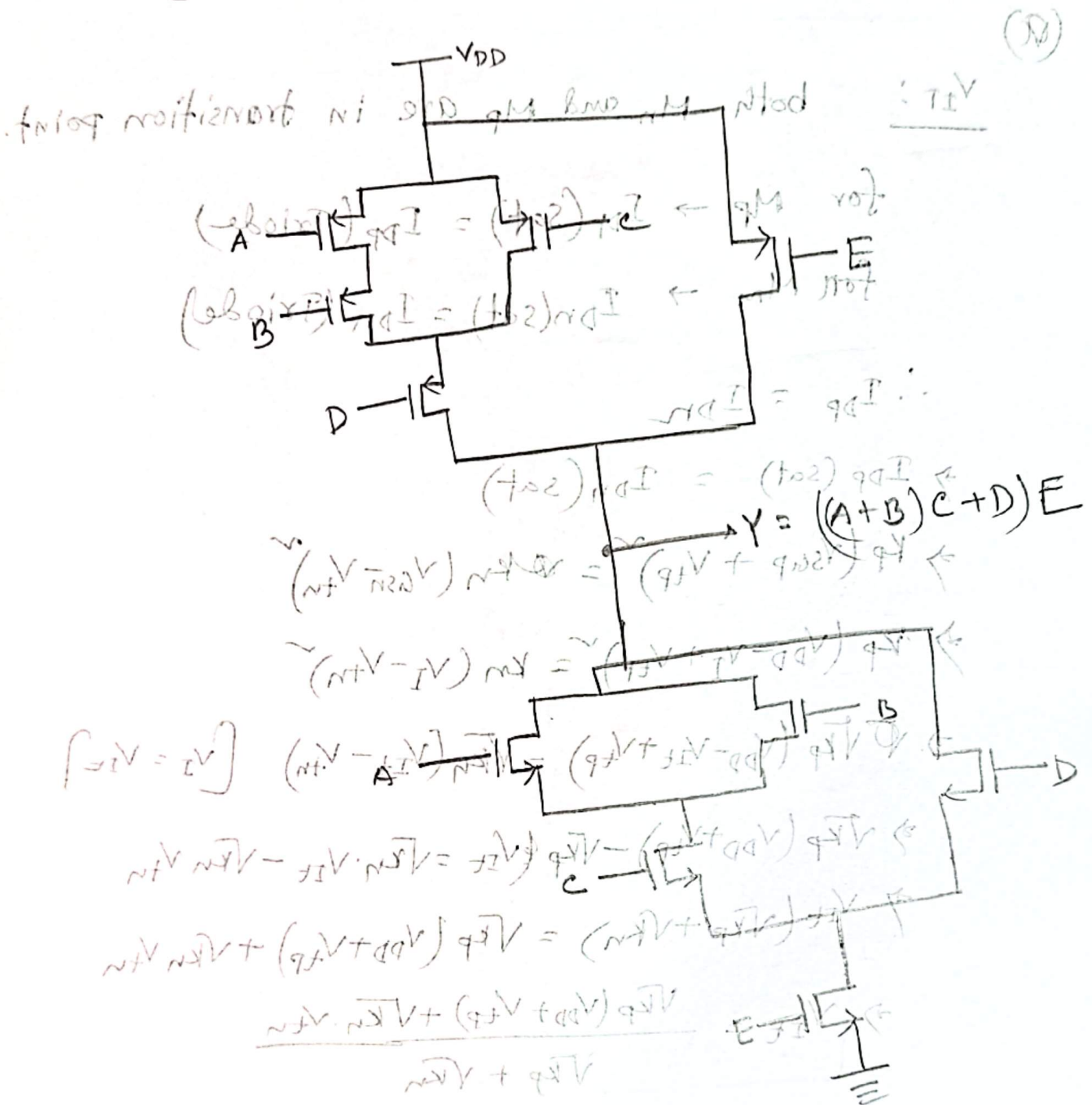
(ii) $\overline{AB} + C = \overline{A + B} + C = \overline{A \cdot B \cdot \overline{C}}$



(iii) $(AB + C)D$



(✓) $\overline{((A+B)C+D)E}$ J-OR map out of m7



Logic realization in CMOS using 7.1V

At low point $V_{in} = 0V$

$V_{out} = V_{DD} = 7.1V$

$V_{in} = 7.1V$

$V_{out} = 0V$